Curriculum Vitae

Dr Michael Lydeamore
Department of Econometrics and Business Statistics
Monash University
Email: michael.lydeamore@monash.edu

APPOINTMENTS HELD

Senior Lecturer - Econometrics and Business Statistics *2021-*
Monash University

COVID-19 Modelling & Forecasting Lead *2020*
Manager, Analytics
Department of Health and Human Services, Victoria

Postdoctoral Research Fellow *2019-2021*
Monash University

Honorary Team Member *2019-*
SaferCare Victoria

Honorary Research Fellow *2019-*
Alfred Health

Research Fellow *2018-2019*
The Kirby Institute, University of New South Wales

EDUCATION

Doctor of Philosophy - Applied Mathematics *2015-2019*
The University of Melbourne
Thesis title: Mechanistic and statistical models of skin disease transmission

Masters of Philosophy - Applied Mathematics *2013-2014*
The University of Adelaide
Thesis title: Approximations of stochastic household models for comparing antiviral allocation schemes
Awarded the Applied Probability Trust prize for the highest mark with a project in Applied Mathematics and Statistics

Bachelor of Mathematical Sciences *2010-2013*
The University of Adelaide
Graduated with a double major in applied and pure mathematics

PUBLICATIONS

PREPRINTS

1. Gamage, J. P., Cook, D., Harrison, P., **Lydeamore, M.**, & Talagala, T. S. (2025). *quollr: An R Package for Visualizing 2-D Models from Nonlinear Dimension Reductions in High-Dimensional Space* (arXiv:2512.18166). arXiv. https://doi.org/10.48550/arXiv.2512.18166

2. Gamage, J. P., Cook, D., Harrison, P., **Lydeamore, M.**, & Talagala, T. S. (2025). *cardinalR: Generating Interesting High-Dimensional Data Structures* (arXiv:2512.18172). arXiv. https://doi.org/10.48550/arXiv.2512.18172

3. Gamage, J. P., Cook, D., Harrison, P., & **Lydeamore, M.** (2025). *Stop Lying to Me: New Visual Tools to Choose the Most Honest Nonlinear Dimension Reduction* (arXiv:2506.22051). arXiv. https://doi.org/10.48550/arXiv.2506.22051

4. Miller, C. M., **Lydeamore, M. J.**, Waddle, A. W., Berger, L., Skerratt, L. F., Flegg, J. A., & Campbell, P. T. (2025). *Mathematical modelling of chytridiomycosis transmission in frogs* (arXiv:2503.06846). arXiv. https://doi.org/10.48550/arXiv.2503.06846

5. Conway, E., Walker, C., **Lydeamore, M.**, Golding, N., Ryan, G., Mavec, D., Oates, J., Kabashima, G., Price, D. J., Shearer, F., Cromer, D., Davenport, M. P., McCaw, J., Eriksson, E. M., Hodgkin, P. D., Wu, L., Le, T. P., Baker, C. M., Mueller, I., & McVernon, J. (2024). *Optimal timing of booster doses in a highly vaccinated population with minimal natural exposure to COVID-19*. medRxiv. https://doi.org/10.1101/2024.05.14.24307386

PEER-REVIEWED ARTICLES

1. **Lydeamore, M.**, Zachreson, C., Conway, E., Shearer, F. M., Baker, C. M., Ross, J. V., Miller, J., McCaw, J. M., Geard, N. L., McVernon, J., & Price, D. J. (2026). Border quarantine, vaccination and public health measures to mitigate the impact of COVID-19 importations in Australia: a modelling study. *Journal of The Royal Society Interface*, *23*(235), 20250144. https://doi.org/10.1098/rsif.2025.0144

2. Tierney, N., Saraswati, C., Babu, A., **Lydeamore, M.**, & Golding, N. (2026). conmat: generate synthetic contact matrices for a given age-stratified population. *Journal of Open Source Software*, *11*(118), 8326. https://doi.org/10.21105/joss.08326

3. Nguyen, Q., **Lydeamore, M. J.**, Dessauer, A., Snell, G. I., Westall, G. P., Peleg, A. Y., & Gardiner, B. J. (2026). Evaluation of Global Immune Biomarkers to Predict Infections in Lung Transplant Recipients. *Clinical Transplantation*, *40*(8), e70654. https://doi.org/10.1111/ctr.70654

4. Le, T. P., Conway, E., Akpan, E., Abell, I. R., Abraham, P., Baker, C. M., Campbell, P. T., Cromer, D., **Lydeamore, M. J.**, McDonough, Y., Mueller, I., Ryan, G., Walker, C., Wang, Y., Carvalho, N., & McVernon, J. (2025). Health impact and cost-effectiveness of COVID-19 booster vaccination strategies in the early post-Omicron era: a dynamic modelling study. *BMJ Global Health*, *10*(9). https://doi.org/10.1136/bmjgh-2024-016096

5. Hao, T., Ryan, G. E., **Lydeamore, M. J.**, Cromer, D., Wood, J. G., McVernon, J., McCaw, J. M., Shearer, F. M., & Golding, N. (2025). Predicting immune protection against outcomes of infectious disease from population-level effectiveness data with application to COVID-19. *Vaccine*, *55*, 126987. https://doi.org/10.1016/j.vaccine.2025.126987

6. **Lydeamore, M. J.**, Wu, D., Donker, T., Gorrie, C., Higgs, C. K., Easton, M., Hennessy, D., Geard, N., Howden, B. P., Cooper, B. S., Wilson, A., Peleg, A. Y., & Stewardson, A. J. (2024). Changes in isolation guidelines for CPE patients results in only mild reduction in required hospital beds. *Infection, Disease & Health*. https://doi.org/10.1016/j.idh.2024.10.004

7. **Lydeamore, M. J.**, Donker, T., Wu, D., Gorrie, C., Turner, A., Easton, M., Hennessy, D., Geard, N., Howden, B. P., Cooper, B. S., Wilson, A., Peleg, A. Y., & Stewardson, A. J. (2024). Carbapenemase-producing enterobacterales colonisation status does not lead to more frequent admissions: a linked patient study. *Antimicrobial Resistance & Infection Control*, *13*(1), 82. https://doi.org/10.1186/s13756-024-01437-x

8. Shearer, F. M., Edwards, L., Kirk, M., Eales, O., Golding, N., Hassall, J., Liu, B., **Lydeamore, M.**, Miller, C., Moss, R., Price, D. J., Ryan, G. E., Sheena, S., Tobin, R., Ward, K., Kaldor, J., Cheng, A. C., Wood, J., & McCaw, J. M. (2024). Opportunities to strengthen respiratory virus surveillance systems in Australia: lessons learned from the COVID-19 response. *Communicable Diseases Intelligence*, *48*. https://doi.org/10.33321/cdi.2024.48.47

9. Shearer, F. M., McCaw, J. M., Ryan, G. E., Hao, T., Tierney, N. J., **Lydeamore, M. J.**, Wu, L., Ward, K., Ellis, S., Wood, J., McVernon, J., & Golding, N. (2024). Estimating the impact of test–trace–isolate–quarantine systems on SARS-CoV-2 transmission in Australia. *Epidemics*, *47*, 100764. https://doi.org/10.1016/j.epidem.2024.100764

10. Le, T. P., Abell, I., Conway, E., Campbell, P. T., Hogan, A. B., **Lydeamore, M. J.**, McVernon, J., Mueller, I., Walker, C. R., & Baker, C. M. (2024). Modelling the impact of hybrid immunity on future COVID-19 epidemic waves. *BMC Infectious Diseases*, *24*(1), 407. https://doi.org/10.1186/s12879-024-09282-4

11. Conway, E., Walker, C. R., Baker, C., **Lydeamore, M. J.**, Ryan, G. E., Campbell, T., Miller, J. C., Rebuli, N., Yeung, M., Kabashima, G., Geard, N., Wood, J., McCaw, J. M., McVernon, J., Golding, N., Price, D. J., & Shearer, F. M. (2023). COVID-19 vaccine coverage targets to inform reopening plans in a low incidence setting. *Proceedings of the Royal Society B: Biological Sciences*, *290*(2005), 20231437. https://doi.org/10.1098/rspb.2023.1437

12. Mitchell, B. G., Stewardson, A. J., Kerr, L., Ferguson, J. K., Curtis, S., Busija, L., **Lydeamore, M. J.**, Graham, K., & Russo, P. L. (2023). The incidence of nosocomial bloodstream infection and urinary tract infection in Australian hospitals before and during the COVID-19 pandemic: an interrupted time series study. *Antimicrobial Resistance & Infection Control*, *12*(1), 61. https://doi.org/10.1186/s13756-023-01268-2

13. Zachreson, C., Shearer, F. M., Price, D. J., **Lydeamore, M. J.**, McVernon, J., McCaw, J., & Geard, N. (2022). COVID-19 in low-tolerance border quarantine systems: Impact of the Delta variant of SARS-CoV-2. *Science Advances*, *8*(14), eabm3624. https://doi.org/10.1126/sciadv.abm3624

14. **Lydeamore, M. J.**, Mitchell, B. G., Bucknall, T., Cheng, A. C., Russo, P. L., & Stewardson, A. J. (2022). Burden of five healthcare associated infections in Australia. *Antimicrobial Resistance & Infection Control*, *11*(1), 69. https://doi.org/10.1186/s13756-022-01109-8

15. Trauer, J. M., **Lydeamore, M. J.**, Dalton, G. W., Pilcher, D., Meehan, M. T., McBryde, E. S., Cheng, A. C., Sutton, B., & Ragonnet, R. (2021). Understanding how Victoria, Australia gained control of its second COVID-19 wave. *Nature Communications*, *12*(1), 6266. https://doi.org/10.1038/s41467-021-26558-4

16. McMahon, J. H., **Lydeamore, M. J.**, & Stewardson, A. J. (2021). Bringing evidence from press release to the clinic in the era of COVID-19. *Journal of Antimicrobial Chemotherapy*, *76*(3), 547–549. https://doi.org/10.1093/jac/dkaa506

17. Zachreson, C., Mitchell, L., **Lydeamore, M. J.**, Rebuli, N., Tomko, M., & Geard, N. (2021). Risk mapping for COVID-19 outbreaks in Australia using mobility data. *Journal of The Royal Society Interface*, *18*(174), 20200657. https://doi.org/10.1098/rsif.2020.0657

18. Sullivan, S. G., Brotherton, J. M., Lynch, B. M., Cheung, A., **Lydeamore, M.**, Stevenson, M., Firestone, S., Canevari, J., Nguyen, H. N. J., & Carville, K. S. (2021). Population-based analysis of the epidemiological features of COVID-19 epidemics in Victoria, Australia, January 2020-March 2021, and their suppression through comprehensive control strategies. *Lancet Regional Health - Western Pacific*, *17*.

19. **Lydeamore, M. J.** (2021). Mathematical models to support Victoria's COVID-19 response: a blunt instrument to a complex problem. *Journal of the Australian Mathematical Society*. https://austms.org.au/wp-content/uploads/2021/07/Lydeamore.pdf

20. **Lydeamore, M. J.**, Campbell, P. T., Price, D. J., Wu, Y., Marcato, A. J., Cuningham, W., Carapetis, J. R., Andrews, R. M., McDonald, M. I., McVernon, J., Tong, S. Y. C., & McCaw, J. M. (2020). Estimation of the force of infection and infectious period of skin sores in remote Australian communities using interval-censored data. *PLOS Computational Biology*, *16*(10), e1007838. https://doi.org/10.1371/journal.pcbi.1007838

SOFTWARE

R PACKAGES

blockstrap *CRAN*

Sample dataframes by group, in the form of a 'block bootstrap'. Entire groups are returned allowing for a single 'observation' to span multiple rows of the dataframe.
Coauthors: Michael Lydeamore, Cash Looi, Kenyon Ng, Mitchell O'Hara-Wild

cardinalR *CRAN*

A collection of functions to generate a large variety of
structures in high dimensions. These data structures are useful for
testing, validating, and improving algorithms used in dimensionality
reduction, clustering, machine learning, and visualization.
Coauthors: Jayani P. Gamage, Dianne Cook, Paul Harrison, Michael Lydeamore, Thiyanga S. Talagala

condensr *CRAN*

Helps automate 'Quarto' website creation for small academic groups.
Builds a database-like structure of people, projects and publications, linking
them together with a string-based ID system. Then, provides functions to automate
production of clean markdown for these structures, and in-built CSS formatting
using CSS flexbox.
Coauthors: Michael Lydeamore

HospitalNetwork *CRAN*

Set of tools to help interested researchers to build hospital networks
from data on hospitalized patients transferred between hospitals. Methods provided
have been used in Donker T, Wallinga J, Grundmann H. (2010) <doi:10.1371/journal.pcbi.1000715>, and Nekkab N, Crépey P, Astagneau P, Opatowski L, Temime L. (2020) <doi:10.1038/s41598-020-71212-6>.
Coauthors: Pascal Crépey, Tjibbe Donker, Clément Massonnaud, Michael Lydeamore

mapycusmaximus *CRAN*

Focus-glue-context (FGC) fisheye transformations to two-dimensional coordinates and spatial vector geometries. Implements a smooth radial distortion that enlarges a focal region, transitions through a glue ring, and preserves outside context. Methods build on generalized fisheye views and focus+context mapping. For more details see Furnas (1986) <doi:10.1145/22339.22342>, Furnas (2006) <doi:10.1145/1124772.1124921> and Yamamoto et al. (2009) <doi:10.1145/1653771.1653788>.
Coauthors: Alex Nguyen, Dianne Cook, Michael Lydeamore

quollr *CRAN*

To construct a model in 2-D space from 2-D nonlinear
dimension reduction data and then lift it to the high-dimensional
space. Additionally, provides tools to visualise the model overlay the
data in 2-D and high-dimensional space. Furthermore, provides
summaries and diagnostics to evaluate the nonlinear dimension
reduction layout.
Coauthors: Jayani P. Gamage, Dianne Cook, Paul Harrison, Michael Lydeamore, Thiyanga S. Talagala

rjsf *CRAN*

An R interface to Jump-Switch-Flow (JSF), a hybrid
simulation framework for compartmental reaction models that combines
stochastic jump dynamics at low population sizes with deterministic flow
dynamics at high population sizes. Simulations are performed using the
'jsf' backend for 'Python' through 'reticulate', with structured R outputs
for inspection, summary, and plotting. The underlying methodology is
described in Germano et al. (2026)
<doi:10.1007/s00285-026-02409-y>.
Coauthors: Thanh Long Vu, Domenic Germano, Michael Lydeamore

spiderorchid *GitHub*

Download and wrangle publication data for Monash EBS academic staff
Coauthors: Rob Hyndman, Michael Lydeamore, Sherry Tee, Parnika Khattri

VS CODE EXTENSIONS

Code Comment Viewer *Open VSX*

Show all HTML comments in the current file in a sidebar and highlight them in the editor.
Version 0.1.3

Collab Comments *Open VSX*

Track comments with author and timestamp for QMD and other files
Version 1.1.0

Quarto: Open Rendered File *Open VSX*

Open an existing rendered output for a Quarto document without rendering it again.
Version 1.1.2

FUNDING

Almost all of this funding was obtained in partnership with large, collaborative teams.

SPARKLE *2024*

Department of Foreign Affairs and Trade
Awarded value: $5,000,000

SHIELD: Surveillance of Healthcare-associated Infections for Effective Local Data *2025*

NHMRC
Awarded value: $2,000,000

Modelling Work in Low and Middle Income Counties in the Western Pacific Region for the COVID-19 SAGE Working Group *2024*

World Health Organisation
Awarded value: $8,157

Modelling To Support Australia's Resilience to and Preparedness for Omicron And Future Sars-Cov-2 Variants *2024*

Commonwealth Government of Australia
Awarded value: $80,059

Contractual Partner to Conduct Research on Healthcare-Associated Infection Rates in Phillippine Hospitals *2023*

World Health Organisation Phillippines
Awarded value: $72,078

Centre of Western Public Health Unit *2023*
Contract Research
Awarded value: $25,000

Extending and comparing methods for projecting social contact matrices *2022*
SPECTRUM/SPARK Seed Funding
Awarded value: $19,438

Quantifying longitudinal relationships between community mobility and COVID-19 case incidence in west metropolitan Melbourne *2022*
SPECTRUM/SPARK Seed Funding
Awarded value: $19,896

Modelling to support Australia's national plan for COVID-19 *2021*
Commonwealth Government of Australia
Awarded value: $147,536

Modelling to support Australia's transition to 'COVID-normal' *2021*
Australian Office of Health Protection
Awarded value: $25,620

CONFERENCE PRESENTATIONS AND CONTRIBUTED TALKS

Gamifying data visualisation: Teaching ggplot2 through competitive code golf *2026*
International Congress on Teaching Statistics

Gamifying data visualisation: Teaching ggplot2 through competitive code golf *2026*
useR! Conference

Gamifying data visualisation: Teaching ggplot2 through competitive code golf *2026*
Australia and New Zealand Industrial Applied Mathematics Conference

Modelling chytridiomycosis transmission in frogs *2025*
Australia and New Zealand Industrial Applied Mathematics Conference

airpurifyr: Open Air Quality Data in R *2024*
WOMBAT Annual Meeting: Open the world with open source

Networks of networks in infectious diseases modelling - Panelist *2024*
Infectious Diseases Modelling Conference

Generating synthetic contact matrices using open-source data *2024*
Australia and New Zealand Industrial Applied Mathematics Conference

Data-Driven Insights into Healthcare Challenges: Two Case Studies - Invited Speaker *2023*
Universitätsklinikum Freiburg Seminar Series

Data-Driven Insights into Healthcare Challenges: Two Case Studies *2023*
UNSW Australia Statistics & Data Science Seminar

Burden of healthcare associated infections in Australia *2022*
Australia and New Zealand Industrial Applied Mathematics Conference

Exponential Random Graph Models and CPE transmission - Invited Speaker *2021*
Melbourne Mathematical Biology Seminar Series

Mathematical modelling for COVID-19 in Victoria, Australia - Invited Speaker *2021*
Math for Industry Forum, Vietnam

The associations between transmission of CPE and ward connectivity: a network analysis *2021*
Australasian Society for Infectious Diseases Annual Scientific Meeting

The burden of healthcare acquired infections in Australian public hospitals *2021*
Australasian Society for Infectious Diseases Annual Scientific Meeting

Decision Making and Mathematical Biology - Victoria's usage of COVID-19 modelling - Invited Speaker *2021*
Mathematical Biology Special Interest Group Workshop

Modelling between-household effective contact and the elimination of COVID-19 in Melbourne, Australia *2021*
Australia and New Zealand Industrial Applied Mathematics Conference

Mathematical modelling and Victoria's response to COVID-19 - Invited Speaker *2020*
Victorian ANZIAM Branch Meeting

Australia's experience and the role of modelling in its responses to COVID-19 - Invited Speaker *2020*
Usher Institute COVID-19 Webinar

Estimating epidemiological quantities for skin sores in remote Australian communities using interval-censored data *2019*
Australia and New Zealand Industrial Applied Mathematics Conference

Estimating epidemiological quantities for skin sores in remote Australian communities using interval-censored data *2018*
NSW-ACT ANZIAM Branch Meeting

Coupled models of Group A Streptococcus and Scabies: How likely is eradication? *2018*
PRISM International Conference

Investigating the dynamics of coupled epidemiological transmission models with application to Group A Streptococcus and Scabies *2018*
Society for Mathematical Biology Annual Meeting

Investigating the dynamics of coupled models with applications to Group A Streptococcus and Scabies *2018*
Australia and New Zealand Industrial Applied Mathematics Conference

Quantifying the age of first infection with skin sores in five remote Australian Aboriginal communities *2017*
Lancefield International Symposium on Streptococci and Streptococcal Diseases

Investigating the dynamics of coupled models with applications to Group A Streptococcus and Scabies *2017*
PRISM Annual Knowledge Transfer and Training Conference

Constructing mathematical models of Group A Streptococcus and Scabies in remote Australian Indigenous communities - Invited Speaker *2017*
Zeeman Institute Seminar Series

Developing a model for the transmission and treatment dynamics of scabies infections a high prevalence setting *2017*
Melbourne-Manchester Joint Workshop

Developing a model for the transmission of Group A Streptococcus *2017*
Australia and New Zealand Industrial Applied Mathematics Conference

Determining the age of first infection from incomplete data *2016*
Modelling Emerging Infections and Neglected Tropical Diseases Workshop

Investigating Intervention Intervals for Scabies Infections *2016*
Australia and New Zealand Industrial Applied Mathematics Conference

TEACHING

LECTURER

[ETC5513] **Reproducible and Collaborative practices** - Chief Examiner *2024-*

[ETC5523] **Communicating with Data** - Chief Examiner *2023-*

[ETO5513] **Reproducible and Collaborative practices** - Chief Examiner *2024-*

[ETC5512] **Wild-Caught Data** *2022-2023*

[ETC5521] **Exploratory Data Analysis** *2022*

TUTOR

[MAST10016] **Mathematics for Biomedicine** *2016-2019*

[MAST30001] **Stochastic Modelling** *2018*

TEACHING SERVICE

[M6036] **Masters of Health Data Analytics** - Course Management Committee *2024-*

AWARDS

Fellow of the Higher Education Academy *2024*

SPECTRUM Annual Meeting - Best Presentation *2024*

University of Melbourne Faculty of Engineering and Information Technology Excellence Award in Interdisciplinary Research *2021*

IPAA Spirit of Service Awards - Finalist *2021*

Engagement Australia Excellence Awards - Outstanding Engagement for Research Impact - Finalist *2021*

Top Poster Award - European Congress of Clinical Microbiology & Infectious Diseases *2021*

IPAA Victoria Leadership in the Public Sector Awards - Finalist *2021*

Best PhD Student Presentation - PRISM² Annual Conference *2017*

Applied Probability Trust Prize - for best Applied Mathematics Postgraduate Thesis *2016*

Australian Postgraduate Award *2015*

Dean's Commendation for Thesis Excellence *2015*

ACADEMIC SERVICE & COMMUNITY ENGAGEMENT

Equity Diversity & Social Inclusion Committee *2024-*
Department representative

International Conference on Prevention and Infection Control *2025*
Abstract Reviewer

SPECTRUM Annual Meeting *2024*
Organising committee

SPARK Short Course in Mathematical Diseases Modelling - OUCRU *2023*
Facilitator

International Conference on Prevention and Infection Control *2023*
Abstract Reviewer

Maths in Industry Study Group - NSW Health *2023*
Moderator

WOMBAT Communicating with Data Workshop *2023*
Organising Committee

Australian and New Zealand Industrial and Applied Mathematics Executive Committee *2022-*
Treasurer

Australian and New Zealand Industrial and Applied Mathematics Executive Committee *2021-2022*
Early Career Representative

SPECTRUM-SPARK Early Career Researcher Committee *2022-*
Chair

SPARK Short Course in Mathematical Diseases Modelling - Mahidol University *2022*
Facilitator

Research Tools Workshop in R *2022*
Facilitator

Mathematical Biology Special Interest Group *2019-2022*
Treasurer

ANZIAM 2021 Conference *2020-2021*
Organising Committee (Treasurer)

Computational Biology Research Initiative - The University of Melbourne *2017*
Postgraduate Representative

Computational Biology Postgraduate Sports Group - The University of Melbourne *2017-2018*
Secretary

Maths in Industry Study Group *2017, 2018, 2019*
Participant

The ConocoPhillips Science Experience - The University of Melbourne *2016-2018*
Activity Organiser

Mathematicians in Schools - CSIRO *2016-2018*
Project Supervisor/Mentor

Reviewer
PLoS Computational Biology; Scientific Reports; Journal of Applied Mathematics; Journal of Statistical Software; Epidemiology & Infection; Mathematics; Medical Journal of Australia; International Tropical Health; Journal of Antimicrobial Resistance; The R Journal

PROFESSIONAL MEMBERSHIPS

Australian and New Zealand Industrial and Applied Mathematics *2015-*
Including the Mathematical Biology Special Interest Group

Australian Mathematics Society *2015-*

Statistical Society of Australia *2024-*